DMF DENTAL TRAINING CENTER: FINAL COACHING

Pharmacology and Anesthesiology

1. Rx in the prescription means
   a. Radiograph - XR / Radiotherapy (RT)
   b. Patient - Pt
   c. Drug
   d. Recipe
2. All are parts of a prescription, except one. Which is it?
   a. Superscription
   b. Inscription
   c. Subscription
   d. Signa
   e. Suprascription
3. Instruction to the patient which means after meals
   a. Pc - Post cibum
   b. Po - Per orem (By mouth)
   c. Pr - Per rectum (rectally)
   d. Ac - Ante cibum - Before meals
4. Qid means
   a. Once a day - OD
   b. Twice a day - Bid
   c. Three times a day - Tid
   d. Four times a day
5. A 68-year-old woman with osteoporosis presents for extraction of a non-restorable mandibular molar. Review of her medications reveals long-term use of a drug known to increase the risk of medication-related osteonecrosis of the jaw (MRONJ). Which medication is most likely responsible?
   a. Amlodipine - CCB - anti HTN
   b. Zoledronic acid → Bisphosphonate (anti-resorptive) drug ← for osteoporosis, AE: MRONJ
   c. Atorvastatin - anti-lipid
   d. Metformin - Biguanides (anti-diabetic)
6. Physostigmine and Neostigmine are classified as _____
   a. Adrenergic-receptor agonist
   b. Cholinergic-receptor agonist
   c. Anticholinesterase → for myasthenia gravis; ↑ Ach; Reversal agent of Atropine
   d. Cholinesterase stimulant
7. A 10-year-old boy diagnosed with ADHD is referred for dental treatment. Which of the following is considered a first-line pharmacologic treatment for ADHD?

Tx: Stimulant

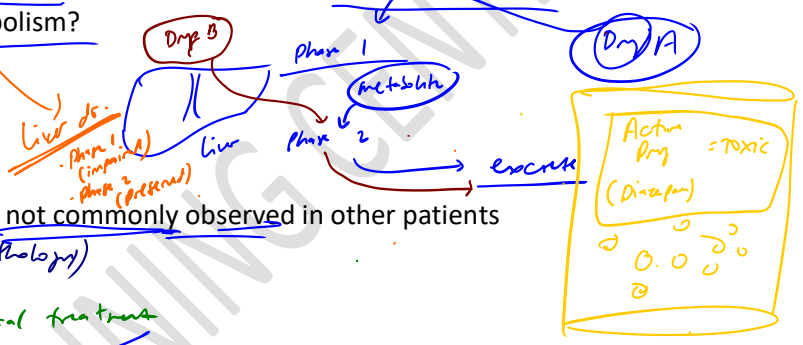
Tach — Focus

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a. Amphetamine (Adderall) — CNS stimulant, indirect-acting adrenergic agent — ADHD, Narcolepsy
b. Guaifenesin – expectorant
c. Biguanide – Metformin (antidiabetic)
d. Loratadine – H_1 Blocker = non-drowsy

8. This is the science of the preparation and dispensing of medicinal drugs.
   a. Pharmacology – Branch of medicine concerned w/ drugs (anything)
   b. Pharmacy
   c. Pharmacokinetics – Branch of Pharmacology (movement of drugs) – ADME
   d. Pharmacodynamics – " " " (effects & mechanism)

9. An elderly patient with severe liver impairment requires anxiolytic therapy before a dental procedure. Which benzodiazepine is preferred because it undergoes direct conjugation and does not require oxidative hepatic metabolism?
   a. Oxazepam
   b. Diazepam
   c. Alprazolam
   d. Midazolam



10. It is an unusual response to a drug that is not commonly observed in other patients
    a. Idiopathic – Unknown cause (pathology)
    b. Idiosyncrasy
    c. Iatrogenic – dz. due to medical treatment
    d. Iatrosyncrasy

11. A patient with localized aggressive periodontitis undergoes scaling and root planing. The periodontist places Arestin® microspheres into the periodontal pockets. What is the active ingredient of Arestin®?
    a. Minocycline
    b. Doxycycline
    c. Penicillin V
    d. Amoxicillin

Tetracycline — periodontitis; Amoxicillin; Metronidazole

12. Among the following drugs, which can be used for xerostomia?
    a. Norepinephrine – adrenergic drug – vasoconstriction
    b. Isoproterenol – adrenergic drug – Bronchodilator
    c. Propranolol – Beta-Blocker – anti-arrhythmic (CI: asthma)
    d. Pilocarpine – Cholinergic drug – ↑ Parasympathetic

13. This is the enzyme inhibited by Pyridostigmine
    a. Monoamine oxidase — MAO Inhibitor
    b. Carbonic anhydrase
    c. Acetylcholinesterase
    d. Protease

14. A dentist decides to prescribe metronidazole as an adjunct to periodontal therapy in a patient with aggressive periodontitis. Which regimen is most commonly recommended among the following choices?
    a. Metronidazole 250 mg BID for 14 days
    b. Metronidazole 250 mg BID ~~for 7 days~~ — TID
    c. Metronidazole 250 mg BID for 5 days

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d. Metronidazole 250 mg BID for 3 days

15. Among the following Adrenergic-receptor agonists, which is used for patients with asthma?
    a. Beta-1 receptor agonist
    b. Beta-2 receptor agonist
    c. Alpha-1 receptor agonist
    d. Alpha-2 receptor agonist
16. It is the prototype of beta-blockers
    a. Propanolol
    b. Metoprolol
    c. Esmolol
    d. Acebutolol
17. Subantimicrobial-dose tetracyclines are sometimes used in periodontal therapy because of their host-modulating effects. Which drug is known to inhibit collagenase activity and matrix metalloproteinases (MMPs)?
    a. Amoxicillin
    b. Minocycline
    c. Metronidazole
    d. Erythromycin
18. Which is not immunosuppressant agent?
    a. Methotrexate
    b. Cyclosporin
    c. Corticosteroids
    d. Tolbutamide
19. They are considered as the most potent and most efficacious diuretic agents
    a. Furosemide
    b. Methazolamide
    c. Thiazides
    d. Mannitol
20. It is a correct statement with regards to the effects of atropine
    a. Atropine increases GI peristalsis
    b. Atropine cannot cross the blood-brain barrier
    c. Atropine increases secretion of saliva and sweat
    d. Atropine at moderate to high doses can increase heart rate
21. A patient presents with acute streptococcal pharyngitis confirmed by culture. Which medication remains the drug of choice for treatment?
    a. Penicillin
    b. Fluconazole
    c. Metronidazole
    d. Acyclovir
22. Sildenafil (Viagra) is used for which of the following conditions?
    a. Asthma
    b. Depression
    c. Hyperglycemia
    d. Hypertension

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23. The following are calcium channel blockers except one. Which is it?
    a. Verapamil
    b. Nifedipine
    c. Imipenem
    d. Diltiazem

24. A dental student is reviewing inhalation sedation agents. Which statement regarding nitrous oxide is MOST accurate?
    a. It is the most potent inhalational anesthetic.
    b. It produces profound surgical anesthesia when used alone
    c. It is a weak anesthetic and requires special delivery armamentarium.
    d. It causes significant cardiovascular depression.

25. A dentist prescribes a 0.12% chlorhexidine gluconate mouth rinse following periodontal therapy. Chlorhexidine is best classified as a:
    a. Antiseptic
    b. Disinfectant only
    c. Antifungal agent
    d. Local anesthetic

26. It is a highly poisonous stimulant drug that is commonly used as a pesticide
    a. Ecstasy
    b. Dextromethamphetamine
    c. LSD
    d. Strychnine

27. Drugs which have the suffix (-navir) is classified as what kind of drug?
    a. Beta-Blockers
    b. Antidepressant
    c. H2-blockers
    d. Protease inhibitors

28. Among the following inhalational anesthetic agents, which is considered the weakest anesthetic based on minimum alveolar concentration (MAC) and anesthetic potency?
    a. Halothane
    b. Nitrous oxide
    c. Isoflurane
    d. Sevoflurane

29. Under Philippine prescribing regulations, why is it important to indicate the generic name of a drug in a prescription?
    a. To increase the cost of treatment
    b. To promote a specific pharmaceutical company
    c. To ensure proper drug identification and allow dispensing of equivalent products
    d. To shorten the prescription

30. All are xanthine stimulants except:
    a. Theophylline
    b. Theobromine
    c. Thioxanthene
    d. Caffeine

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31. Why should a dentist include his/her name and professional license number in a prescription?
    a. To advertise the dental clinic
    b. To increase patient confidence only
    c. To qualify for insurance reimbursement
    d. To establish legal accountability and validate the prescription

32. Tardive dyskinesia is a common side effect of which of the following drugs?
    a. Antihypertensive
    b. Antipsychotics
    c. Diuretics
    d. Analgesics

33. A patient reports a history of severe allergy to amoxicillin. Which drug is most likely to exhibit cross-allergenicity?
    a. Azithromycin
    b. Cephalexin
    c. Clindamycin
    d. Metronidazole

34. When inducing general anesthesia, the effects of the general anesthetic agent is seen to least affect which of the?
    a. Cerebrum
    b. Cerebellum
    c. Medulla
    d. Midbrain

35. 1st generation antipsychotic is ~~not un~~commonly associated with extrapyramidal syndromes. Example of this class of antipsychotic is clozapine.
    a. Both statements are true
    b. Both statements are false
    c. The first statement is true, the second is false
    d. The first statement is false, the second is true

36. Levodopa, amantadine and carbidopa are drugs used to treat which of the following conditions?
    a. Depression
    b. Parkinsonism
    c. Schizophrenia
    d. Bipolar disorder

37. Which antibiotic should generally be avoided during the first trimester of pregnancy because of its potential adverse effects on fetal development?
    a. Penicillin V
    b. Amoxicillin
    c. Tetracycline
    d. Cephalexin

38. Among the following drugs, which may interfere with the antiplatelet activity of aspirin?
    a. Penicillin
    b. Ibuprofen
    c. Synthetic glucocorticoids
    d. Amlodipine

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39. A patient presents with an acute dentoalveolar abscess of pulpal origin. Which antibiotic remains the drug of choice when antibiotic therapy is indicated?
    a. Erythromycin
    b. Tetracycline
    c. Penicillin V
    d. Metronidazole

40. All are antidepressant drugs except:
    a. MAO inhibitors
    b. SSRI and SNRI
    c. Tricyclic antidepressants
    d. Strychnine

41. A 4-year-old child accidentally ingests a large amount of fluoride toothpaste and develops signs of acute fluoride toxicity. Which immediate treatment is most appropriate?
    a. Activated charcoal
    b. Naloxone
    c. Milk or calcium-containing products
    d. Vitamin K

42. This is the primary source of antibiotics we are currently using in medicine
    a. Fungus and viruses
    b. Viruses and mycoplasmas
    c. Mycoplasma and rickettsia
    d. Bacteria and fungus

43. Quinolone is derived from which of the following agents?
    a. Quinine
    b. Quinidine
    c. Triamcinolone
    d. Ceftriaxone

44. The following are ~~not true~~ statements regarding Clindamycin and only one is ~~not false~~. Which is ~~not~~ an ~~un~~true statement that describes Clindamycin?
    a. It is a cell wall synthesis inhibitor
    b. It doesn't penetrate the bone tissue well
    c. It is usually given together with erythromycin for more effective treatment
    d. It is effective against gram-negative bacteria only
    e. Its use is primarily for anaerobic infections

45. Why is 2% lidocaine with 1:100,000 epinephrine considered the gold standard local anesthetic formulation in dentistry?
    a. It has the shortest duration of action.
    b. It is the least toxic anesthetic available.
    c. It provides an excellent balance of safety, anesthesia, and hemostasis.
    d. It is effective without nerve block techniques.

46. It is a barbiturate that is commonly used as a maintenance drug to prevent epileptic seizures
    a. Phenobarbital
    b. Thiopental
    c. Amobarbital

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d. Secobarbital

47. Which of the following is a major advantage of nitrous oxide-oxygen inhalation sedation?
    a. Produces deep general anesthesia
    b. Causes prolonged postoperative recovery
    c. Rapid onset and rapid recovery with titratable effects
    d. Eliminates the need for local anesthesia
48. Paradoxical excitement is seen in which of the following drugs
    a. Antipsychotics
    b. Barbiturates
    c. Benzodiazepines
    d. Analgesics
49. Among are the different general anesthetic inhalation agents. Which is the prototype?
    a. Sevoflurane
    b. Halothane
    c. Desflurane
    d. Nitrous oxide
    e. Isoflurane
50. A patient with advanced liver cirrhosis requires dental treatment under local anesthesia. Which anesthetic is preferred because it is primarily metabolized by plasma esterases?
    a. Lidocaine
    b. Mepivacaine
    c. Bupivacaine
    d. Articaine
51. It is considered as the most potent agent among the inhalation anesthetics
    a. Sevoflurane
    b. Halothane
    c. Desflurane
    d. Nitrous oxide
    e. Isoflurane
52. Digitoxin is a drug effective for patients with heart failure because of which of the following?
    a. It increases the force of contraction of the heart
    b. It is a diuretic agent
    c. It constricts the vessels
    d. It reduces the rate of contraction of the ventricles
53. This is the drug used as an antidote for malignant hyperthermia
    a. Dantrolene
    b. Midazolam
    c. Succinylcholine
    d. Ketamine
54. A patient with an odontogenic infection reports a history of immediate hypersensitivity reaction to penicillin. Which antibiotic is commonly recommended as an alternative?
    a. Penicillin V
    b. Amoxicillin
    c. Clindamycin

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d. Ampicillin

55. Which drug is not used to achieve neuroleptic analgesia?
    a. Droperidol
    b. Nitrous oxide
    c. Fentanyl
    d. None of the above

56. All are classified as beta-lactam antibiotics except one. Which is the exception?
    a. Penicillin
    b. Azithromycin — Macrolides (Inhibits 50S Ribosome)
    c. Cephalosporin
    d. Monobactam
    e. Carbapenem

57. This is a type of penicillin that is also known as anti-staphylococcal penicillin
    a. Broad-spectrum penicillin - anti-pseudomonal penicillin (Piperacillin/Ticarcillin)
    b. Penicillinase-resistant penicillin — greatly resistant to staphylococcal penicillinase.
    c. Aminopenicillin (amoxicillin/ampicillin)
    d. Narrow-spectrum penicillin - natural penicillin (Pen G/Pen VK)

1st (2nd gen) (Dental infections)
1st Broad ⊕⊕⊕
5th Broader ⊖⊖⊖

58. The following are statements regarding cephalosporins. Which of the following is true?
    a. The first generation cephalosporins have spectrum of action predominantly against gram negative organisms
    b. The first generation is a broad-spectrum antibiotic while higher generations have broader-spectrum.
    c. Higher generations of cephalosporins have increasing activity against gram positive organisms
    d. First generation is a broad-spectrum antibiotic while higher generations have decreasing spectrum.

59. Which is classified as a 2nd generation cephalosporin?
    a. Cefazolin — 1st
    b. Cefalexin — 1st
    c. Cefotaxime — 3rd
    d. Cefuroxime

60. The following are included in Aminoglycoside class of antibiotic except one. Which is it?
    a. Streptomycin
    b. Aminomycin
    c. Gentamicin
    d. Tobramycin

5 min

61. The most common side effect associated with tetracycline is _____
    a. Bone and teeth anomalies
    b. Nausea and vomiting
    c. Red man syndrome → Vancomycin
    d. Black hairy tongue - any broad-spectrum (long-term)

62. The type of ribosome that is inhibited by tetracycline is_____
    a. 30s
    b. 50s

Aminoglycosides

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c. 70s
d. 80s

63. The mechanism of action of NSAIDs are _____
a. They act primarily on lipoxygenase enzyme
b. They inhibit the action of cyclooxygenase
c. They inhibit prostaglandin function
d. They promote prostaglandin breakdown

64. An antibiotic known to cause aplastic anemia
a. Tetracycline
b. Piperacillin
c. Imipenem
d. Chloramphenicol

65. Which of the following is a true statement regarding antibiotics?
a. Bacteriostatic and bactericidal agents have synergistic drug interaction
b. Bacteriostatic and bactericidal agents have additive drug interaction
c. Bacteriostatic and bactericidal agents have antagonistic drug interaction
d. Bacteriostatic and bactericidal agents have potentiation drug interaction

66. This is the most common route used when giving aminoglycosides
a. IV
b. IM
c. Oral
d. Transdermal

67. The ratio of the median lethal dose (LD50) to the median effective dose (ED50) is the _____
a. Morbidity index
b. Mortality index
c. Anesthetic ratio
d. Therapeutic index

68. Peripheral neuropathy is seen in patients given nitrous oxide and isoniazid. This is due to vitamin B9 deficiency
a. Both statements are true
b. Both statements are false
c. The first statement is true, the second is false
d. The first statement is false, the second is true

69. Which of the following is an anti-metabolite?
a. 5-fluorouracil
b. Abraxine
c. Oxaliplatin
d. Vinblastine

70. This is a drug commonly used for patients with hyperthyroidism
a. Propylthiouracil
b. Levothyroxine
c. Liothyronine
d. Liotrix

71. Sulfonylureas, Biguanides and Thiazolidinediones are used for _____

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a. Cancers
b. Diabetes mellitus
c. Asthma
d. Adrenal disorders

72. This is the oldest type of NSAID
a. Ibuprofen
b. Mefenamic acid
c. Aspirin
d. Acetaminophen

73. All are byproducts of the cyclooxygenase pathway except one. Which is it?
a. Leukotrienes
b. Prostaglandins
c. Prostacyclin
d. Thromboxane A2

74. A drug commonly associated with Reye's syndrome
a. Acetaminophen
b. Aspirin
c. Synthetic glucocorticoids
d. Methimazole

75. Among the following non-narcotic analgesics, which doesn't have any anti-inflammatory effects and not associated with GI irritation?
a. Sulindac
b. Celecoxib
c. Acetaminophen
d. Indomethacin

76. Which is not immunosuppressant agent?
a. Methotrexate
b. Cyclosporin
c. Corticosteroids
d. Tolbutamide

77. All are laxatives except one. Which is the exception?
a. Psyllium
b. Lactulose
c. Bisacodyl
d. Loperamide
e. Sorbitol

78. This is a drug that is used for patients with high lipid profile
a. Atorvastatin
b. Sulfonylureas
c. Metformin
d. Bismuth subsalicylate

79. Generics act of 1988
a. RA 9165
b. RA 6675

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c. RA 6657
d. RA 9615

80. This is the mechanism of action of bacteriostatic antibiotic, sulfonamide
    a. It prevents the microorganism from using oxygen
    b. It disrupts the cellular membrane
    c. It competes with PABA in folic acid synthesis
    d. It coagulates bacterial proteins
81. This is an example of an alpha-2 receptor agonist agent
    a. Methyldopa → anti-HTN
    b. Phenylephrine – vasoconstrictor / nasal decongestants
    c. Prazosin – α1 blocker
    d. Albuterol – "Salbutamol" = β2 agonist
82. These are NSAIDs generally safe for patients with GI diseases
    a. Mefenamic acid
    b. Ibuprofen
    c. Cox-2 inhibitors
    d. Aspirin
83. Cocaine is the very first local anesthesia and it is also the only naturally occurring. Cocaine and Tetracaine are both vasoconstrictors.
    a. Both statements are true
    b. Both statements are false
    c. The first statement is true, the second is false
    d. The first statement is false, the second is true
84. Epinephrine and Levonordefrin are added to local anesthetics because _____.
    a. They increase the potency of the LA
    b. Of their vasoconstrictive properties
    c. They decrease the risk of allergies to the LA
    d. Of their ability to decrease pain during LA administration
85. Among the following types of nerve fibers, which is the most difficult to anesthetize?
    a. Small unmyelinated fibers
    b. Small myelinated fibers
    c. Large unmyelinated fibers
    d. Large myelinated fibers
86. It is a type of pain that has no organic basis and is fixed upon some portion of the anatomy is referred to as _____
    a. Referred pain → pain on another part of body
    b. Psychogenic pain → mental factor
    c. Phantom pain
    d. Intractable pain → not easily managed by analgesics
87. An infection makes the local environment alkaline. An alkaline pH results to more ionized form of local anesthetics causing them to be less effective
    a. Both statements are true
    b. Both statements are false
    c. The first statement is true, the second is false

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d. The first statement is false, the second is true

88. Among the following local anesthetics, which has the fastest onset of action?
    a. Mepivacaine
    b. Lidocaine
    c. Articaine
    d. Bupivacaine

89. For an adult patient the maximum dose of lidocaine with epinephrine is _____ and the total daily dose must not go beyond _____
    a. 7mg/kg and 500 mg
    b. 7mg/kg and 200 mg
    c. 4.5mg/kg and 300 mg
    d. 4.5mg/kg and 500 mg

90. How many mg of mepivacaine will be received by the patient if you gave him 2 cartridges of 3% mepivacaine (1.8ml).
    a. 36mg
    b. 54mg
    c. 60mg
    d. 108mg

91. A 2% lidocaine with 1:100,000 epinephrine contains how much lidocaine and epinephrine?
    a. 3.6 mg lidocaine and 0.018mg epinephrine
    b. 36mg lidocaine and 0.18 mg epinephrine
    c. 36 mg lidocaine and 0.018 mg epinephrine
    d. 3.6 mg lidocaine and 0.18 mg epinephrine

92. Supplementary injection when doing RCT for a mandibular molar with irreversible pulpitis.
    a. Local infiltration
    b. Field block
    c. Mandibular blocking
    d. Intrapulpal

93. This is the local anesthesia that is specifically made for dental use
    a. Lidocaine
    b. Mepivacaine
    c. Bupivacaine
    d. Articaine

94. For the local anesthetics, for every 1% solution there is _____
    a. 0.10mg/ml of anesthetic
    b. 1mg/ml of anesthetic
    c. 10mg/ml of anesthetic
    d. 100mg/ml of anesthetic

95. This is the maximum number of epinephrine-containing cartridges that can be given for cardiovascular compromised patients.
    a. 1cc of 1:50:000 epinephrine
    b. 2cc of 1:50:000 epinephrine
    c. 1cc of 1:100,000 epinephrine

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d. 2cc of 1:100,000 epinephrine

96. This is an anti-microbial content of local anesthetics that also causes allergies to patients
    a. Bisulfites
    b. Methylparaben
    c. Amide
    d. Ester

97. These are the nerves responsible for transmitting fast sharp pain
    a. A-Beta fibers
    b. A-Delta fiber
    c. C Fibers
    d. B fibers

98. The most common cause of a high plasma level of local anesthetics following a dental injection is:
    a. Inadequate hepatic function
    b. Inadequate renal function
    c. Depositing in an excessive concentration
    d. Intravascular injection

99. Correct statements about procaine include all of the following, except one. Which is it?
    a. It has an ester linkage
    b. It is relatively long-acting
    c. It interferes with sodium influx during deposition
    d. Its metabolic product can inhibit the action of sulfonamide

100. When anesthetic solution is deposited in close proximity to small nerve terminal nerve branches, the technique is referred to as:
    a. Infiltration
    b. Field block
    c. Topical
    d. Nerve block

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